

CRED-RAPTOR

A Hierarchical RAG-based LLM Pipeline for Corporate Credit Risk
Assessment

Extended Abstract Presentation • CADE 2026

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The Bottleneck: Qualitative Blind Spots

Theoretical Motivation

- ✓ **Quantitative Limitations:** Financial KPIs (liquidity, leverage) provide snapshots but fail to capture "soft" risk signals.
- ✓ **Qualitative Value:** Research confirms factors like management experience and reliability are the most valued in credit decisioning.
- ✓ **Operational Burden:** Risk officers manually parse 100+ page filings, leading to inconsistency and delayed credit decisions.

Key Regulatory Challenges

- ⚠ **Black-Box Prohibitions:** Modern banking regulations demand auditable reasoning meaning AI cannot be a "black box."
- ⚠ **Feature Traceability:** Extracted metrics must be grounded in specific source document paragraphs for audit compliance.

System Architecture

Our pipeline integrates three distinct feature sets into a unified **XGBoost** predictor, balancing computational efficiency with nuanced depth:

1. Quantitative Layer

Automated calculation of core financial KPIs from SEC 10-K data. Provides the objective baseline for borrower capacity.

2. Qualitative Layer

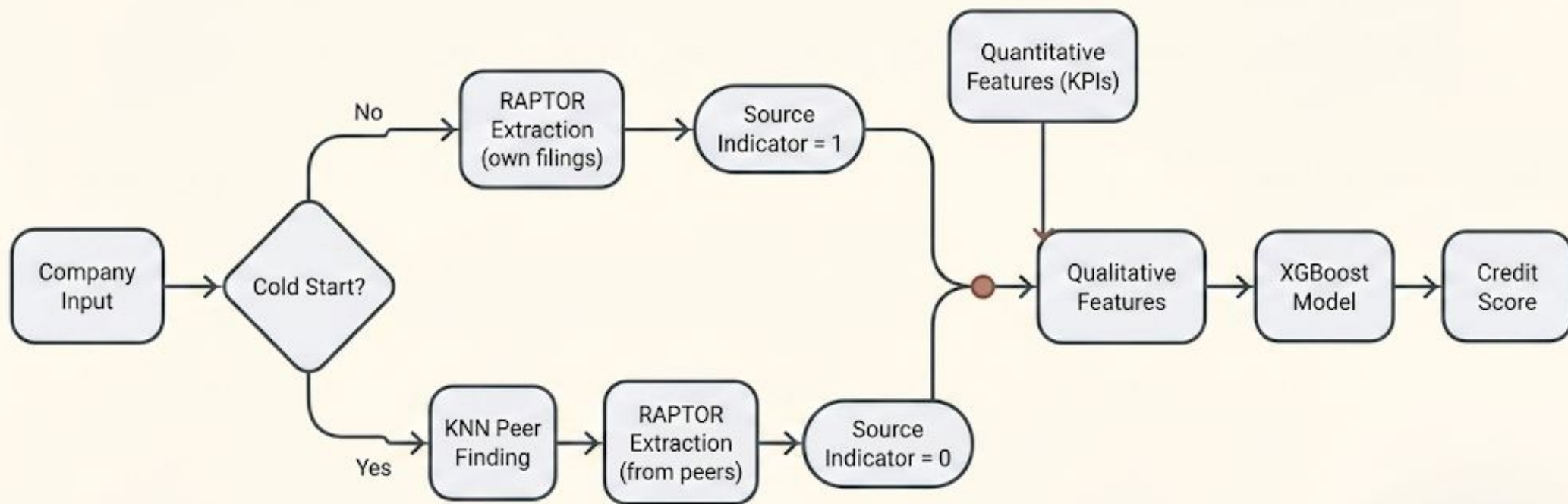
Scored dimensions extracted via **RAPTOR**. This captures business insights and operational execution risks.

3. Source Indicator

A binary flag (0/1) for cold-start companies. Ensures the model distinguishes between self-filings and peer-proxy data.

Architectural Defensibility

By using XGBoost for the final prediction instead of a direct LLM end-to-end approach, we maintain statistical rigor and interpretability while leveraging LLMs solely for high-dimensional feature extraction.



Quantitative KPIs

Three programmatic indicators calculated directly from structured SEC filings to represent financial health:

Profit Margin: $(\text{Net Income} / \text{Net Sales}) \times 100$. Measures operational efficiency and pricing power.

Return on Assets (ROA): $(\text{Net Income} / \text{Total Assets}) \times 100$. Assesses effectiveness of firm's asset utilization.

Debt Ratio: $(\text{Long-Term Debt} / \text{Total Assets}) \times 100$.

Data Integrity: Computed only when denominators are non-zero; rounded to two decimal places for uniformity.

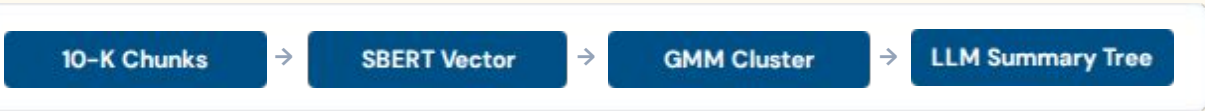
Qualitative Engine: RAPTOR Mechanism

IMPLEMENTATION & MULTI-SCALE CONTEXT JUSTIFICATION

⚙️ Technical Pipeline Execution

- **Document Segmentation:** Lengthy 10-K filings are programmatically parsed into contiguous text chunks to maintain baseline semantic local context.
- **SBERT Embedding:** Chunks are transformed into high-dimensional vectors to capture precise financial and lexical semantics.
- **Gaussian Mixture Models (GMM):** Implements soft clustering on vector spaces, continuing recursively to build a multi-layered hierarchical tree architecture.
- **Recursive Summarization:** Nodes within identical clusters are summarized via an LLM, re-embedded, and re-clustered until terminal root convergence is achieved.

RATOR Ingestion Workflow



Why RAPTOR Over Flat RAG?

Standard flat chunking creates information silos. Credit risk factors are systemic narratives spanning dozens of pages. RAPTOR preserves macro-thematic context alongside micro-details.

Mathematical Choice: GMM Soft Clustering

Unlike rigid partition algorithms (e.g., K-Means), GMMs operate probabilistically. This allows a single risk factor such as a complex legal dispute to belong to both *Regulatory Risk* and *Revenue Stability Risk* clusters simultaneously.

Deterministic Output Grounding

The 5 risk dimensions are scored on a strict 0–10 scale. To safeguard the downstream XGBoost architecture against API volatility, the template forces **neutral fallback metrics** if structural extraction anomalies or ambiguities arise.

Qualitative Features

The system evaluates **5 Credit Risk Dimensions** on a 10-point scale. LLM prompts are injected with context retrieved from the RAPTOR tree:

Dimension	Scientific Analysis Goal
Industry & Business	Competitive positioning and sector-specific cyclical risks.
Revenue Stability	Assessment of recurring vs. volatile income streams in the 10-K.
Operational & Asset	Supply chain resilience and quality of the underlying asset base.
Management Execution	Strategy alignment and historical performance vs. leadership guidance.
Regulatory & Licensing	Legal exposure, pending litigations, and compliance status.

Cold-Start: KNN Peer Identification

The Challenge

New or private companies lack historical SEC filings, creating a "qualitative vacuum" that breaks standard RAG pipelines.

- 🟡 **Information Vacuum:** Insufficient textual records for direct RAPTOR extraction.
- 🟡 **Feature Incompleteness:** Narrative risk signals cannot be generated for the model.
- 🟡 **Scoring Failure:** Models default to "blind" baseline scores, ignoring strategic context.

The Solution

- ✓ **KNN Peer Identification:** Finds peers based on asset size, debt, and sector similarity.
- ✓ **Proxy Text Extraction:** RAPTOR runs hierarchical retrieval on peer filing text.
- ✓ **Feature Imputation:** Peer group qualitative insights serve as the target's baseline.
- ✓ **Provenance Tracking:** Sets Source Indicator = 0 to flag proxy-derived data.

Experimental Baselines

Baseline model

Input: Quantitative Features Only.

An ordinary **XGBoost model** fed solely with financial KPIs extracted from SEC Filings.

Represents the standard industry approach for automated credit scoring.

CRED-RAPTOR

Input: Financial KPIs + Qualitative Risk Dimensions

Our approach adds multi-dimensional qualitative insights extracted from **RAPTOR** to same XGBoost model.

Captures soft risk signals typically requiring manual expert review.

Future Research Directions

As an **Extended Abstract**, CRED-RAPTOR lays the architectural foundation for several critical expansions:

Domain-Specific LLMs

Fine-tuning models on specialized financial datasets (credit memos) to improve qualitative nuance and sentiment detection.

Synthetic Labeling

Using LLMs to synthesize credit score labels where ground-truth is proprietary or sparse, aiding in zero-shot learning.

Multi-Stage Caching

Optimizing compute costs and latency by caching RAPTOR trees, allowing instant audit re-traces during regulatory reviews.

"Ongoing experimentation focuses on data augmentation to enrich cold-start profiles and validating the pipeline against private banking datasets."

Questions?

Thank you for your time.

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